Rigorous Proof Architecture for the Erdős-Straus Conjecture

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Abstract

This document presents a structured framework and partial resolution towards the Erdős-Straus Conjecture. It defines the formal axiomatic boundaries, establishes the contextual algebraic structures, details zero-ellipse lemmas for modular reduction, and formulates an architecture ready for automated formalization in systems such as Lean 4.

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1 Axiomatic Definitions and Type Specifications

Let $\mathbb{N}$ denote the set of positive integers $\{1, 2, 3, \ldots\}$.

Definition 1.1 (Erdős-Straus Equation). *For $n \in \mathbb{N}_{\geq 2}$, a solution to the Erdős-Straus equation is an ordered triplet $(x, y, z) \in \mathbb{N}^3$ such that:*

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

In polynomial form, this is equivalent to finding positive integers x, y, z satisfying:

$$4xyz = n(xy + yz + zx)$$

2 Contextual Literature Research

The Erdős-Straus conjecture is fundamentally a problem of Diophantine equations over the rationals. Key results from the literature include:

- **Mordell's Theorem:** Bounds on the number of solutions to Diophantine equations of degree 3.
- **Webb and Schinzel (1983):** Demonstrated that the conjecture holds for all n except possibly those in certain congruence classes modulo 840.
- **Elsholtz and Tao (2013):** Established upper bounds on the number of solutions to the equation $4/n = 1/x + 1/y + 1/z$.

An analogy can be drawn to the weakly resolved Erdős-Graham conjecture, where similar modular constraints dictate the density of subset sums. Recent studies by authors such as Miguel Angel Lopez have classified solutions into types, such as Type A and Type B, by defining a complete congruence system. Furthermore, Philemon Urbain Mballa has explored an unexpected connection between the discrete zeta function and the Erdős-Straus conjecture through additive decomposition.

3 Proof Strategy and Lemmas

We proceed by modular reduction, examining the conjecture for prime n. If the conjecture holds for all primes, it holds for all integers by a simple scaling argument.

Lemma 3.1 (Prime Reduction Lemma). *If the Erdős-Straus equation has a solution for all prime numbers $p \geq 2$, then it has a solution for all integers $n \geq 2$.*

Proof. Let $n = p \cdot k$, where p is prime and $k \in \mathbb{N}$. Assume there exists a solution for p:

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

Dividing both sides by k yields:

$$\frac{4}{pk} = \frac{1}{ak} + \frac{1}{bk} + \frac{1}{ck}$$

Since $n = pk$, we have:

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

where $x = ak$, $y = bk$, and $z = ck$. This completes the proof of the reduction. □

Lemma 3.2 (Polynomial Parameterization for $p \equiv 3 \pmod 4$)**.** *For a prime $p \equiv 3 \pmod 4$, there exists a parameterization of solutions.*

Proof. Let $p \equiv 3 \pmod 4$. This implies there exists an integer $k \geq 0$ such that $p = 4k+3$. We set $x = k + 1$. Because $k \geq 0$, we have $x \geq 1$, ensuring $x \in \mathbb{Z}_{>0}$. Note that $x = (4k + 4)/4 = (p + 1)/4$. We evaluate the remaining difference:

$$\begin{aligned}
\frac{4}{p} - \frac{1}{x} &= \frac{4}{p} - \frac{1}{k+1} \\
&= \frac{4(k+1) - p}{p(k+1)} \\
&= \frac{4k + 4 - (4k + 3)}{p(k+1)} \\
&= \frac{1}{p(k+1)}
\end{aligned}$$

We employ the standard Egyptian fraction splitting identity:

$$\frac{1}{A} = \frac{1}{A+1} + \frac{1}{A(A+1)}$$

Applying this to $A = p(k + 1)$, we set: $y = p(k + 1) + 1$ and $z = p(k + 1)(p(k + 1) + 1)$ Since $p \geq 3$ and $k \geq 0$, both y and z are strictly positive integers. By substitution:

$$\begin{aligned}
\frac{1}{y} + \frac{1}{z} &= \frac{1}{p(k+1)+1} + \frac{1}{p(k+1)(p(k+1)+1)} \\
&= \frac{p(k+1)}{p(k+1)(p(k+1)+1)} + \frac{1}{p(k+1)(p(k+1)+1)} \\
&= \frac{p(k+1)+1}{p(k+1)(p(k+1)+1)} \\
&= \frac{1}{p(k+1)}
\end{aligned}$$

Adding $\frac{1}{x} = \frac{1}{k+1}$, we get:

$$\begin{aligned}
\frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= \frac{1}{k+1} + \frac{1}{p(k+1)} \\
&= \frac{p}{p(k+1)} + \frac{1}{p(k+1)} \\
&= \frac{p+1}{p(k+1)} \\
&= \frac{4k+4}{p(k+1)} \\
&= \frac{4(k+1)}{p(k+1)} \\
&= \frac{4}{p}
\end{aligned}$$

This constructs explicitly a positive integer solution for all $p \equiv 3 \pmod 4$. □

4 Architecture for Autoformalization

To facilitate formal verification, we define the structure in a pseudo-Lean 4 syntax block, establishing the required types and theorems.

```
import Mathlib.Data.Nat.Basic
import Mathlib.Tactic.Omega
import Mathlib.Tactic.Ring

namespace ErdosStraus

-- Axiomatic definition of the Erdos-Straus property
def SatisfiesErdosStraus (n : Nat) : Prop :=
  Exists (fun x => Exists (fun y => Exists (fun z => x > 0 /\ y >
     0 /\ z > 0 /\ 4 * x * y * z = n * (y * z + x * z + x * y))))

-- Complete demonstration of Lemma 3.2 based on the document's
   parametrization
lemma erdos_straus_mod_4_3 (k : Nat) : SatisfiesErdosStraus (4 * k
    + 3) := by
  let n := 4 * k + 3
  let x := k + 1
  let y := n * (k + 1) + 1
  let z := n * (k + 1) * (n * (k + 1) + 1)
  use x, y, z
  refine \<by omega, by omega, by omega, ?_\>
  dsimp [x, y, z, n]
  ring

-- General Theorem (Open Conjecture for the set of residual
   classes)
theorem erdos_straus_conjecture (n : Nat) (hn : n >= 2) :
   SatisfiesErdosStraus n := by
  admit

end ErdosStraus
```

References

- Dagnachew Jenber Negash (2018). *Solutions to Diophantine Equation of Erdos-Straus Conjecture.* arXiv:1812.05684v2.

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- Philemon Urbain Mballa (2023). *An Unexpected Connection Between the Discrete Zeta Function and the Erdos-Straus Conjecture Under Mballa's Conjecture*. arXiv:2311.08272v1.